Resampling-Based Model Selection for PE SDFs

How to empirically measure the risk-adjusted performance of non-traded cash flows

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1 Idea

The paper asks a simple empirical question: which stochastic discount factor (SDF) models give the most credible risk adjustment for private-equity cash flows?

The difficulty is that private-equity samples are small, lumpy, and organized by vintage year. A model can look good or bad because of the particular way we group funds into pricing moments. Instead of treating one grouping as fixed, we make the grouping itself part of the empirical design. We repeatedly build pricing moments from different fund partitions, estimate the SDF on some of those moments, and evaluate pricing errors on held-out moments.

The method therefore has two goals:

- **Model selection:** choose the SDF model that gives the lowest holdout pricing error for PE cash flows.
- **Design comparison:** compare whether it is better to build moments from newly resampled within-vintage fund portfolios or from traditional vintage-year portfolios.

The paper is closest in spirit to Korteweg and Nagel [2016] because the estimator works with aggregated pricing moments. It also retains the key DLP12 idea that the moments price private cash flows directly [Driessen et al., 2012]. The new contribution is the resampling layer: instead of committing to one arbitrary construction of moments, we ask which SDFs price many admissible constructions out of sample.

2 Guiding Example

Suppose a candidate SDF has three parameters: an alpha term and two factor loadings. Then we need at least three pricing moments to estimate it. A natural resampling design is therefore:

- build three estimation folds,
- build one validation fold,
- estimate the SDF on the three estimation moments,
- score the fitted SDF on the validation moment.

Figures 1 and 2 show the two benchmark ways to build these folds. In both figures, each dot is one fund and the active vintages are 2005–2010. Vintages 2011–2012 are excluded because they are too young or not sufficiently liquidated.

3 Two Canonical Sampling Schemes

Proposed split: within-vintage fund resampling. In the proposed design, every fold contains funds from every active vintage. This keeps the vintage-year support fixed across estimation and validation moments.

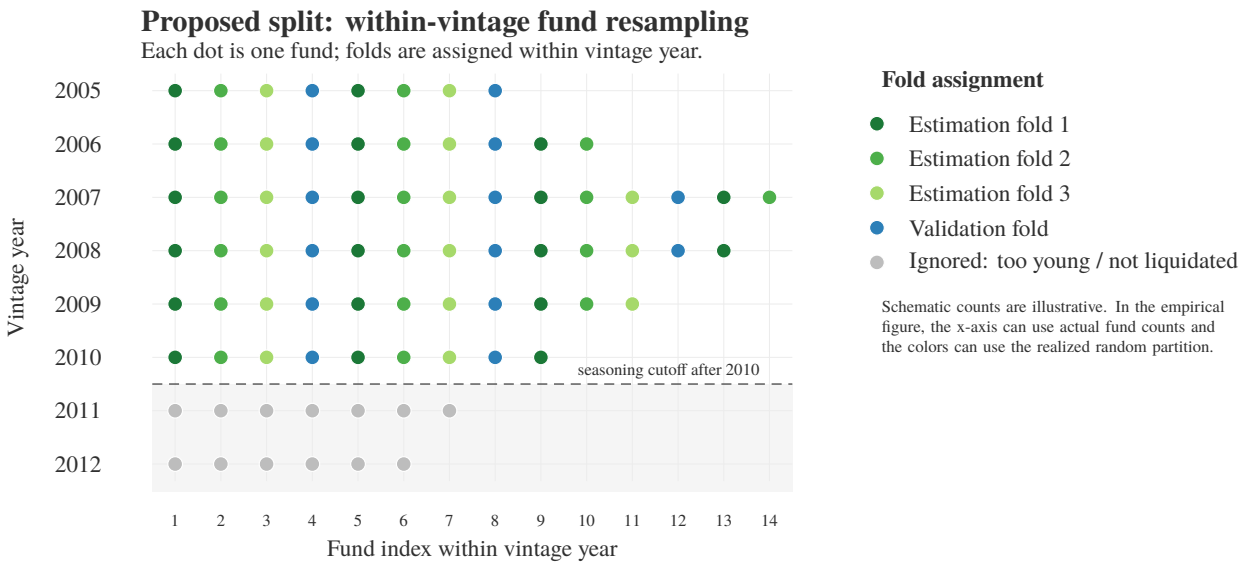

Figure 1: Proposed within-vintage fund resampling scheme. Each active vintage is split into three estimation cells and one validation cell. Fold 1 contains the dark-green funds from every vintage, fold 2 contains the medium-green funds from every vintage, fold 3 contains the light-green funds from every vintage, and the validation fold contains the blue funds from every vintage.

The folds differ because they contain different funds, not because they represent different macro or vintage states. This is the cleanest design for asking whether an SDF prices PE cash flows robustly across fund-level resamples.

Benchmark split: shuffled vintage-year portfolios. In the traditional design, the primitive object is a vintage-year portfolio. A fold contains whole vintages, not fund sub-samples from all vintages. This is easy to explain and resembles standard PE portfolio construction. But the validation moment now differs from the estimation moments in vintage composition. A failure on the validation fold may therefore reflect poor SDF pricing, or it may reflect extrapolation across vintage states.

The first version of the paper should focus on this contrast. If within-vintage resampling delivers more stable estimates and lower holdout pricing errors, the evidence favors building all-vintage subportfolios before estimating the SDF. If shuffled vintage-year portfolios perform similarly, the simpler vintage-year construction may be sufficient.

4 Objects and Notation

Let $v = 1, \ldots, V$ index active vintage years, and let $\mathcal{I}_v$ be the set of funds in vintage v. Let M denote a candidate SDF model with parameter vector $\theta \in \Theta_M \subset \mathbb{R}^{p(M)}$.

We use P estimation folds and h holdout folds. For the guiding three-parameter example, $P = 3$ and $h = 1$. When comparing models with different numbers of parameters, a clean implementation is to use a common number of estimation folds $P = p_{\max}$, where $p_{\max} = \max_M p(M)$.

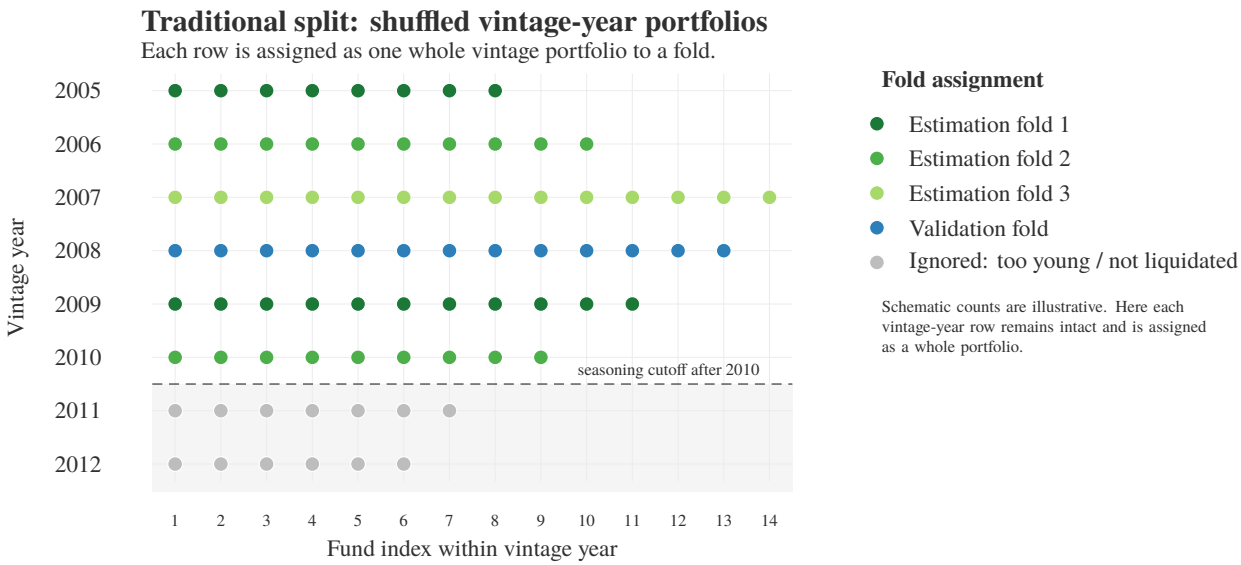

Figure 2: Traditional shuffled vintage-year portfolio scheme. Funds are first aggregated into whole vintage-year portfolios, and whole vintages are then assigned to folds. Here, 2005 and 2009 enter estimation fold 1, 2006 and 2010 enter estimation fold 2, 2007 enters estimation fold 3, and 2008 is the validation fold.

4.1 Within-vintage resampling

For sampled split $n = 1, \ldots, N$, split the funds within each vintage into $P + h$ cells:

$$\mathcal{P}_{n,v} = \{C_{n,v,1}, \ldots, C_{n,v,P+h}\}, \qquad C_{n,v,r} \subset \mathcal{I}_v.$$

The cells are disjoint and cover the funds in vintage v. The rth cross-vintage fold is

$$F_{n,r} = \{C_{n,1,r}, C_{n,2,r}, \ldots, C_{n,V,r}\}.$$

Thus fold r contains one fund sub-sample from every active vintage.

At the single-fund level, the number of admissible assignments is usually too large to enumerate. We therefore draw Monte Carlo samples from a partition rule Π:

$$\mathcal{P}_n \sim \Pi(\cdot \mid P, h, \mathcal{I}_1, \ldots, \mathcal{I}_V).$$

For now, Π should be simple and transparent: random assignment within each vintage, subject to no empty cells and reasonable balance in fund counts or economic size.

4.2 Vintage-year portfolio benchmark

For the benchmark design, first aggregate all funds in each vintage v into a vintage-year portfolio. Then assign whole vintages to folds. The notation is the same at the fold level, but the economic object differs: $F_{n,r}$ now contains whole vintage-year portfolios rather than within-vintage fund cells.

5 Pricing Moments

For a fold $F_{n,r}$, let $\epsilon_{n,v,r}(\theta; M, \eta)$ be the pricing error from the vintage component of that fold. The design choice η collects choices such as the alpha definition, loss function, and cash-flow weighting.

The aggregated fold moment is

$$m_{n,r}(\theta; M, \eta) = \sum_{v \in A_{n,r}} \omega_{n,v,r}\, \epsilon_{n,v,r}(\theta; M, \eta),$$

where $A_{n,r}$ is the set of vintages represented in fold r, and $\omega_{n,v,r}$ are vintage or value weights. Under within-vintage resampling, $A_{n,r}$ is the full active vintage set for every fold. Under shuffled vintage-year portfolios, $A_{n,r}$ is only the set of whole vintages assigned to fold r.

This distinction is the core empirical design choice. The proposed method creates comparable fold moments that all span the same vintage domain. The benchmark creates moments from different vintage domains.

6 Estimation and Holdout Scoring

For each sampled split n, let E_n be the set of estimation folds and H_n the validation fold. The split-specific estimator solves

$$\hat{\theta}_n(M, \eta, \Pi) = \arg\min_{\theta \in \Theta_M} \frac{1}{P} \sum_{j \in E_n} L\big(m_{n,j}(\theta; M, \eta)\big)\,.$$

The validation score is the pricing loss on the held-out fold:

$$S_n(M, \eta, \Pi) = L\Big(m_{n,H_n}(\hat{\theta}_n(M, \eta, \Pi); M, \eta)\Big)\,.$$

Aggregate this score across many sampled assignments:

$$\text{Score}_N(M, \eta, \Pi) = \frac{1}{N} \sum_{n=1}^{N} S_n(M, \eta, \Pi).$$

The preferred model-design pair (M, η, Π) is the one with the lowest average holdout pricing loss. Parameter stability is still informative, but the primary ranking criterion should be predictive pricing performance: does the fitted SDF price cash flows that were not used in estimation?

6.1 Choice of Pricing Moment

The pricing error $m_{n,j}(\theta; M, \eta)$ in the loss L is a design choice, not a primitive. Two canonical families appear in the private-capital SDF literature: the *dollar NPV* moment of Driessen et al. [2012] and the *PME-ratio* moment of Kaplan and Schoar [2005], Driessen et al. [2009], and Ang et al. [2018]. Because the resampling-based ranking of (M, η, Π) can in principle be sensitive to this choice, we discuss both here and treat the NPV moment as the baseline, the PME-ratio as a robustness check.

Dollar NPV moment. For fold cell j in split n, aggregate discounted cash flows into one cumulative pricing error,

$$m_{n,j}^{\mathrm{NPV}}(\theta; M, \eta) = \sum_{i \in C_{n,j}} w_i \sum_{t=0}^{T_i} \Psi_{0,t}(\theta; M, \eta)\, CF_{t,i}, \qquad CF_{t,i} = D_{t,i} - X_{t,i},$$

where $X_{t,i}$ are contributions, $D_{t,i}$ distributions (including terminal NAV for seasoned funds), and $\Psi_{0,t}$ the cumulative SDF. At the population parameter, $E[m_{n,j}^{\mathrm{NPV}}(\theta_0)] = 0$: the risk-adjusted NPV of fund cash flows vanishes.

Pros. (i) *Linearity* in cash flows yields tractable GMM derivatives, influence functions, and CLT-based inference. (ii) *Additivity* across funds makes the fold structure of Section 11, as well as subsampling and the bootstrap, transparent: the cell moment is the sum of fund-level moments, and the split moment is the sum of cell moments. (iii) *Compatibility with the Euler NAV restriction* for unseasoned vintages: the one-period quarterly Euler error $\eta_{t,i}(\theta)$, discounted to time zero and summed, telescopes into m^{NPV} when $\mathrm{NAV}_{0,i} = \mathrm{NAV}_{T_i,i} = 0$. The same pricing object is therefore used for seasoned and unseasoned vintages, up to a finite-sample NAV correction that vanishes as funds mature. No denominator can be near zero; there is no ratio instability.

Cons. (i) *Scale-dependence.* A \$1B fund with a 1% relative NPV error contributes ten times more to the moment than a \$100M fund with the same relative error, so efficient GMM weighting implicitly tilts the estimator toward mega-funds. The influence function is mechanically heavy-tailed in the fund-size distribution rather than in genuine pricing disagreement. (ii) *Dollar-denominated comparisons across vintages* with heterogeneous aggregate commitments are not directly meaningful; normalization is left to the econometrician. (iii) *Sign cancellation.* A cross-sectional mean of zero is consistent with systematic under-pricing of one subgroup offsetting over-pricing of another, which the scalar moment cannot detect. This motivates the fine-grained fold structure but does not eliminate the concern within a fold.

PME-ratio moment. Define the risk-adjusted PME of fund i as

$$\mathrm{PME}_i(\theta; M, \eta) = \frac{\sum_t \Psi_{0,t}(\theta; M, \eta)\, D_{t,i}}{\sum_t \Psi_{0,t}(\theta; M, \eta)\, X_{t,i}},$$

from which two natural cell-level moments follow: the *average-PME* form $m_{n,j}^{\mathrm{PME}}(\theta) = |C_{n,j}|^{-1} \sum_{i \in C_{n,j}} \big(\mathrm{PME}_i(\theta) - 1\big)$ and the *pooled-PME* form $m_{n,j}^{\text{PME-pool}}(\theta) = \big(\sum_{i,t} \Psi_{0,t} D_{t,i}\big) / \big(\sum_{i,t} \Psi_{0,t} X_{t,i}\big) - 1$. Both have mean zero at θ_0 under standard regularity, but they estimate different population objects (mean of a ratio versus ratio of means).

Pros. (i) *Scale-invariance.* The ratio is unit-free, so fund-size heterogeneity does not mechanically up-weight mega-funds. A \$100M fund and a \$1B fund enter the cross-sectional moment on equal footing. (ii) *Practitioner interpretability.* PME is the native performance metric of the industry; reporting results in the same units aids dissemination. (iii) *Bounded downside.* The fund-level PME is bounded below by zero, so $\mathrm{PME}_i - 1$ is bounded below by -1: a single disaster fund cannot dominate the moment the way it can dominate a dollar NPV.

Cons. (i) *Nonlinearity.* The moment is a ratio in θ; its derivative involves the ratio of two correlated sums and is numerically fragile when the denominator (discounted contributions) is small. This occurs mechanically for young funds with back-loaded capital calls — precisely the vintages that the Euler NAV restriction is designed to retain. (ii) *Jensen-style bias between forms.* The average-PME and pooled-PME moments can differ in finite samples by a term of order $\mathrm{Var}(X_i)/\bar{X}^2$; the two are asymptotically equivalent only under

degenerate size distributions. Researchers must commit to one and defend it. (iii) *No telescoping with the Euler NAV restriction.* Integrating unseasoned vintages $\mathcal{V}^U$ into a PME framework typically requires treating the latest $\text{NAV}_{T_i,i}$ as a terminal distribution and discounting it over the fund's entire remaining horizon — exactly the long-horizon discounting mechanism that motivates ϵ^U in Section 9. The PME-ratio framework therefore reintroduces the exploding-alpha sensitivity for $\mathcal{V}^U$ that the NPV/Euler combination was designed to eliminate. (iv) *Inference.* Upper-tail skewness in PME_i (funds can return many times their capital, but cannot lose more than it) distorts CLT-based standard errors and t-statistics unless explicitly robustified, e.g., by bootstrapping the cell-level moment or by employing median-based loss functions.

Choice for the empirical implementation. The telescoping compatibility with the Euler NAV restriction and the tractability of inference make the dollar NPV moment our primary specification: it delivers a unified pricing restriction across $\mathcal{V}^S$ and $\mathcal{V}^U$ without introducing long-horizon NAV discounting through the back door. We additionally report two variants as robustness checks: (a) a scale-normalized NPV, $m^{\text{NPV}}_{n,j} / \sum_{i \in C_{n,j}} K_i$ where K_i is committed capital, which preserves linearity while neutralizing the large-fund tilt; and (b) the pooled-PME moment restricted to seasoned vintages $\mathcal{V}^S$, which is where PME is unambiguously well-defined. Stability of the preferred (M, η, Π) across these three moments is the appropriate robustness diagnostic: agreement suggests genuine model fit, while disagreement would flag moment-induced identification.

7 Why Validation Is Needed

If a model has p parameters and we estimate it on exactly p moments, the in-sample fit can look deceptively good. This is especially dangerous in small PE samples because exact identification can hide instability. The validation fold creates a more demanding question: after fitting the model on P moments, does it also price a held-out PE cash-flow portfolio?

The main empirical specifications should be:

- **Main comparison:** $P = p_{\max}$, $h = 1$, so all candidate models are compared on a common fold architecture.
- **Exact-identified diagnostic:** $P = p(M)$, $h = 0$, to study feasibility and parameter instability.
- **Model-specific validation:** $P = p(M)$, $h = 1$, to compare exact-identified estimation plus one holdout fold.
- **Scheme comparison:** run the same model list under within-vintage resampling and shuffled vintage-year portfolios.

8 Alpha Design

The resampling method does not by itself decide what alpha should mean. That choice is central to the paper. The cleanest baseline is to use a cumulative, NPV-style alpha rather than a period-by-period intercept alpha.

8.1 Baseline: cumulative alpha

For fold $F_{n,j}$, let $B_t(\beta)$ be the cumulative benchmark return implied by factor loadings β. A direct-alpha-style fold moment is

$$\psi_{n,j}(a,\beta) = \sum_t \frac{cf_{n,j,t}}{B_t(\beta)} \exp\big(-a\,\tau_{n,j,t}\big),$$

where $cf_{n,j,t}$ is the fold cash flow at time t, and $\tau_{n,j,t}$ is elapsed time from the fold origin to date t. For fixed β, the fold-implied cumulative alpha solves

$$\sum_t \frac{cf_{n,j,t}}{B_t(\beta)} \exp\big(-a_{n,j}(\beta)\,\tau_{n,j,t}\big) = 0.$$

This is close to the KN16/GPME/direct-alpha tradition because alpha is an NPV-style abnormal-performance object [Korteweg and Nagel, 2016, Gredil et al., 2023, Korteweg and Nagel, 2025]. It is also more natural for long-horizon PE cash flows than a monthly intercept alpha.

8.2 Alternative alpha definitions

The paper should still compare alpha definitions:

- **Cumulative alpha:** the recommended baseline, estimated from benchmark-discounted cash flows.
- **Error-term alpha:** alpha as the mean pricing-error component, as in the machine-learning approach of Tausch and Pietz [2024].
- **Intercept alpha:** a DLP12-style robustness specification, useful for comparison but vulnerable to exploding-alpha behavior in small or weakly identified samples [Driessen et al., 2012, 2008].

The empirical question is not only which SDF factors perform best, but also which alpha definition yields stable and predictive PE pricing moments.

9 Euler NAV Restrictions for Unseasoned Vintages

The empirical algorithm in Section 11 excludes insufficiently seasoned vintages. This wastes data from recent vintages that carry useful pricing information. Including them naively—by treating the latest reported NAV as a terminal distribution—amplifies alpha instability, because a single large NAV position is discounted over the fund's entire remaining horizon.

We propose a different treatment. For unseasoned vintages, replace the single cumulative pricing restriction with a sequence of one-period Euler restrictions on quarterly NAV returns. This retains the information from recent vintages while avoiding long-horizon discounting. The resulting estimator is a *NAV-bridged cash-flow estimator*: asymptotically a pure cash-flow estimator, with well-conditioned finite-sample corrections for young funds.

9.1 Seasoned and unseasoned vintage sets

Partition the active vintages into seasoned and unseasoned sets,

$$\mathcal{V} = \mathcal{V}^S \cup \mathcal{V}^U.$$

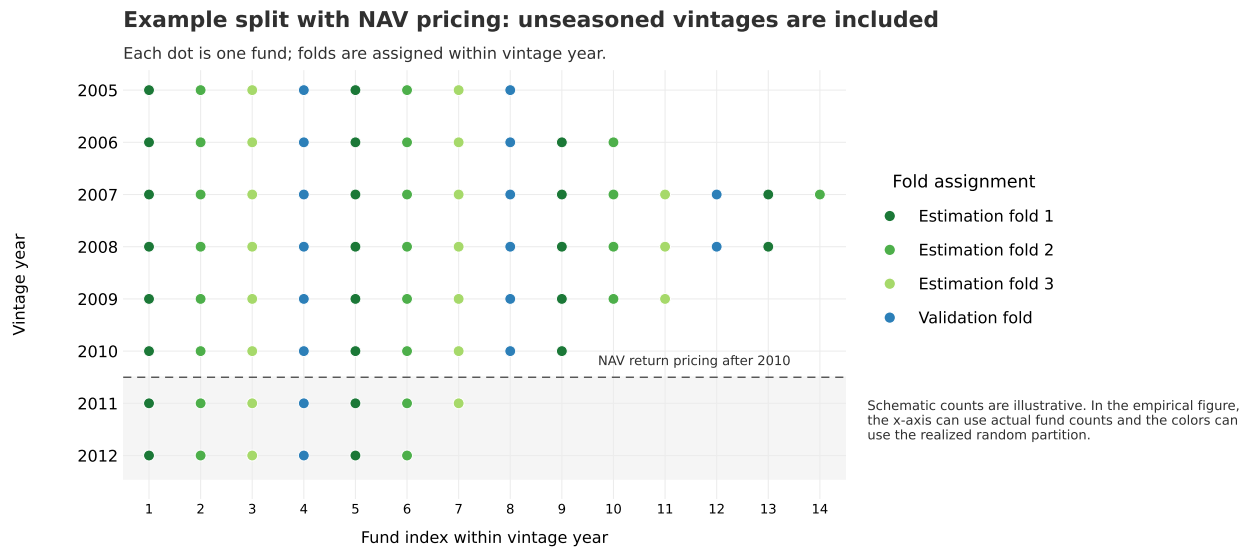

Figure 3: Within-vintage fold assignment under the NAV-bridged cash-flow estimator. Unseasoned vintages 2011–2012 ($\mathcal{V}^U$, grey band) are included in estimation and validation like the seasoned vintages ($\mathcal{V}^S$), but contribute the one-period Euler NAV error ϵ^U in place of the cumulative cash-flow error ϵ^S.

A vintage v is *unseasoned* ($v \in \mathcal{V}^U$) if its aggregate residual NAV exceeds a fraction δ of total committed capital; otherwise it is *seasoned* ($v \in \mathcal{V}^S$). In practice, $\delta \in [0.10, 0.20]$ is a sensible range. For seasoned vintages, the pricing restriction is the standard cumulative cash-flow error as defined in Section 5. For unseasoned vintages, we use the one-period Euler pricing error defined next.

9.2 One-period Euler pricing error

For a fund i in an unseasoned vintage with quarterly NAV observations at dates $t = 0, 1, \ldots, T_i$, define the one-period pricing error for quarter t:

$$\eta_{t,i}(\theta) = \Psi_{t-1,t}(\theta) \cdot \left(CF_{t,i} + \mathrm{NAV}_{t,i}\right) - \mathrm{NAV}_{t-1,i},$$

where $CF_{t,i}$ is the net cash flow during the quarter, $\mathrm{NAV}_{t,i}$ is the reported net asset value at quarter-end, and $\Psi_{t-1,t}(\theta)$ is the one-period stochastic discount factor.

The restriction $E[\eta_{t,i}(\theta_0)] = 0$ is an Euler equation: the beginning-of-period NAV equals the risk-adjusted present value of within-period distributions plus the end-of-period NAV. It is the same object that prices a one-period return on a traded asset, applied here to the NAV-implied gross return of the fund.

For an unseasoned vintage $v \in \mathcal{V}^U$ and fund cell $C_{n,v,r}$, define the cell-level Euler pricing error as the discounted sum of quarterly errors:

$$\epsilon^U_{n,v,r}(\theta) = \sum_{i \in C_{n,v,r}} w_i \sum_{t=1}^{T_v} \Psi_{0,t-1}(\theta)\, \eta_{t,i}(\theta).$$

The cumulative discount factor $\Psi_{0,t-1}(\theta)$ converts each quarterly error to time-zero units, ensuring that ϵ^U is denominated in the same present-value units as the standard cash-flow pricing error ϵ^S for seasoned vintages.

9.3 Telescoping and asymptotic nesting

The discounted Euler errors telescope, which establishes that the Euler approach nests the standard cash-flow approach.

Proposition (Telescoping). *For a fund with inception value* $\mathrm{NAV}_{0,i} = 0$ *and terminal value* $\mathrm{NAV}_{T_i,i} = 0$ *(fully liquidated), the sum of discounted Euler errors equals the standard cumulative pricing error:*

$$\sum_{t=1}^{T_i} \Psi_{0,t-1}(\theta)\, \eta_{t,i}(\theta) = \sum_{t=1}^{T_i} \Psi_{0,t}(\theta)\, CF_{t,i}.$$

Proof. Expand the left-hand side using $\Psi_{0,t-1} \cdot \Psi_{t-1,t} = \Psi_{0,t}$:

$$\begin{aligned}\sum_{t=1}^{T_i} \Psi_{0,t-1}\, \eta_{t,i} &= \sum_{t=1}^{T_i} \left[\Psi_{0,t}\,(CF_{t,i} + \mathrm{NAV}_{t,i}) - \Psi_{0,t-1}\,\mathrm{NAV}_{t-1,i}\right] \\ &= \sum_{t=1}^{T_i} \Psi_{0,t}\, CF_{t,i} \;+\; \underbrace{\sum_{t=1}^{T_i} \left[\Psi_{0,t}\,\mathrm{NAV}_{t,i} - \Psi_{0,t-1}\,\mathrm{NAV}_{t-1,i}\right]}_{\text{telescopes to } \Psi_{0,T_i}\mathrm{NAV}_{T_i} - \mathrm{NAV}_{0,i} \,=\, 0}.\end{aligned}$$

□

This identity has three consequences:

1. **Nesting.** For fully liquidated funds, the Euler pricing error equals the standard cash-flow pricing error. If all vintages are seasoned, the estimator reduces to a pure cash-flow estimator.
2. **Convergence.** As an unseasoned fund matures and its NAV declines toward zero, the Euler error converges continuously to the cash-flow error. There is no discontinuity at the seasoned/unseasoned boundary.
3. **Asymptotic irrelevance.** Only the most recent vintages within a bounded horizon D carry nonzero NAV terms. As $V \to \infty$, the fraction $|\mathcal{V}^U|/V \to 0$. The estimator is therefore *asymptotically a pure cash-flow estimator*; the Euler restrictions are finite-sample corrections that vanish in the limit.

9.4 Why one-period restrictions stabilize alpha

Under the standard cumulative approach, including an unseasoned fund with residual $\mathrm{NAV}_T > 0$ requires discounting by the cumulative SDF $\Psi_{0,T}(\theta)$, which contains the factor $(1+\alpha)^{-T}$. For large T or large NAV_T, a small perturbation in α produces a large swing in the pricing error. This mechanism could aggravate the exploding-alpha issue.

The quarterly Euler restriction replaces this single long-horizon discounting with T one-period discountings, each involving $(1+\alpha)^{-1}$. The sensitivity to α is proportional to the quarter length Δt, not exponential in the fund horizon T. Moreover, each quarterly error $\eta_{t,i}$ subtracts NAV_{t-1} from a discounted quantity of comparable magnitude, so the error is well-conditioned regardless of fund size.

9.5 Integration into fold moments

The fold moment retains its original form:

$$m_{n,r}(\theta; M, \eta) = \sum_{v \in \mathcal{V}^S \cap A_{n,r}} \omega_{n,v,r}\, \epsilon^S_{n,v,r}(\theta; M, \eta) \;+\; \sum_{v \in \mathcal{V}^U \cap A_{n,r}} \omega_{n,v,r}\, \epsilon^U_{n,v,r}(\theta; M, \eta),$$

where ϵ^S is the standard cumulative cash-flow pricing error and ϵ^U is the Euler-based error defined above. Both terms are pricing errors that equal zero under correct pricing. The entire estimation and validation machinery from Sections 5 and 6 applies without modification—only the construction of vintage-level pricing errors differs between seasoned and unseasoned vintages.

10 Relation to DLP12 and KN16

The paper can be positioned in one paragraph:

- **From DLP12:** price private cash flows directly with an SDF.
- **From KN16:** use aggregated pricing moments and an NPV-style abnormal-performance object.
- **New here:** treat the construction of PE pricing moments as a resampling problem and rank SDF models by holdout pricing errors across many admissible fund assignments.
- **Also new:** Euler NAV restrictions for unseasoned vintages that telescope into the standard cash-flow restriction, avoiding both sample waste and exploding-alpha degeneracy (Section 9).

The paper is therefore not a repair of the existing SDF paper. It is a new empirical-design paper: how should we build, resample, and validate PE pricing moments when the sample is small and the moment construction is consequential?

11 Empirical Algorithm

For each candidate model M, alpha design η, and partition scheme Π:

1. Partition vintages into seasoned ($\mathcal{V}^S$) and unseasoned ($\mathcal{V}^U$) sets (Section 9).
2. Choose P estimation folds and h holdout folds.
3. Draw a sampled assignment n from Π.
4. Build fold cash-flow portfolios and fold pricing moments $m_{n,r}(\theta; M, \eta)$.
5. Estimate $\hat{\theta}_n(M, \eta, \Pi)$ on the P estimation folds.
6. Evaluate the fitted model on the holdout fold.
7. Repeat over many sampled assignments.
8. Rank model-design pairs by average holdout pricing loss.

Alongside the mean holdout loss, report diagnostics that help interpret the ranking:

- median and dispersion of holdout losses,
- dispersion of $\hat{\theta}_n$ across assignments,
- frequency of non-convergence,
- frequency of boundary or exploding-alpha solutions,
- sensitivity of results to within-vintage resampling versus shuffled vintage-year portfolios.

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